

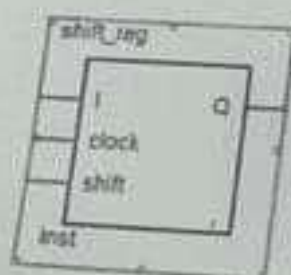
Q2) Draw the logic circuit for the current MODE code

```

LIBRARY IEEE;
USE IEEE.std_logic_1164.all;
USE IEEE.std_logic_arith.all;
ENTITY logic IS
    port (
        a: in std_logic_vector(3 downto 0);
        f: out std_logic_vector(3 downto 0);
    );
END ENTITY logic;
ARCHITECTURE logic OF logic IS
    BEGIN
        with a select
            f <= "0001" when "00",
                "0010" when "01",
                "0100" when "10",
                "1000" when others;
    END

```

Q3) Write the Entity for the following circuit



Entity of flip_flop is

Port (I, shift: in std_logic;
 clock: in std_logic;
 Q: out std_logic);

End of flip_flop;

Decoder 2×2^2
 2×4



A ₁	A ₂	F ₀	F ₁	F ₂	F ₃
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

$$F_0 = \overline{A_1} \overline{A_2}$$

$$F_1 = \overline{A_1} A_2$$

$$F_2 = A_1 \overline{A_2}$$

$$F_3 = A_1 A_2$$

Q4) Write the
 Library
 Use IEEE
 Entity
 Port (

End

Arch

Begin

Proc

Begin

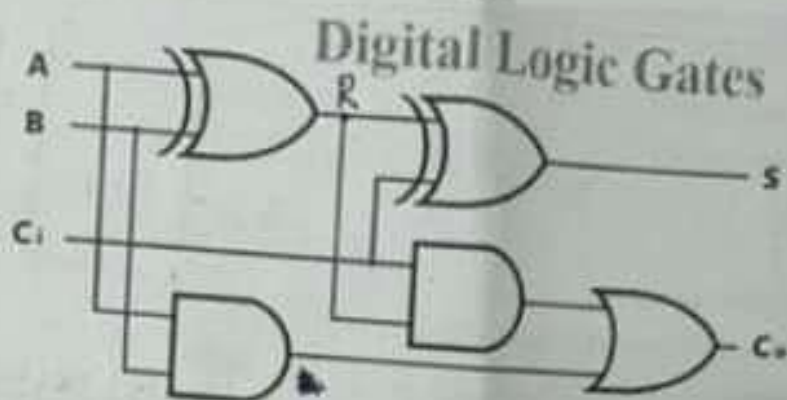
Q<

End

End

د پاره کلاسه

Q1) Write the VHDL statement to the following circuit



~~A~~

$$S \leftarrow (A \text{ XOR } B) \text{ XOR } C_i$$

~~Co~~

$$\text{OR } S \leftarrow R \text{ XOR } C_i$$

$$C_o \leftarrow (A \text{ AND } B) \text{ OR } (R \text{ AND } C_i)$$

✓

$$\begin{aligned} & (A \text{ XOR } B) \text{ XOR } C \\ & (R \text{ AND } C) \\ & (A \text{ AND } B) \text{ OR } (R \text{ AND } C_i) \end{aligned}$$

26
30

2 bit

Q1) Write the VHDL code to describe the 4 to 1 Mux using the Conditional signal assignment.

Library IEEE;

Use IEEE.Std.Logic.StdLogic1164.all;

Entity Mux4x1 is

Port (I₀, I₁, I₂, I₃ : in Std.Logic;

S : in Std.Logic (2 downto 0);

Q : out Std.Logic^{value});

End Mux4x1;

Architecture Mux-behavior of Mux4x1 is

Begin

Process (I₀, I₁, I₂, I₃, S)

Begin

Q <= I₀ when S = "00" else

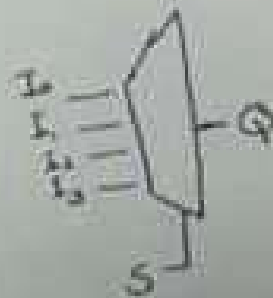
I₁ when S = "01" else

I₂ when S = "10" else

I₃;

End Process;

End Mux-behavior;



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